

OIL AND NATURAL GAS CORPORATION LIMITED (ONGC)

Supply Chain Management at ONGC and Case Study of an associated business process.

Under the guidance of Mr. Ramandeep Singh Sir(ONGC)
(15/05/26 to 19/06/26)

By-
Maitreyi Pandey
Birla Institute of Technology, Mesra
Production and Industrial Engineering

1: Overview:

Oil and Natural gas corporation limited (ONGC) is India's largest crude oil and natural gas organization and one of the leading public sector enterprises in the country. The organization plays a vital role and contributes significantly in meeting the energy demands through exploration, production and supply of natural gas and crude oil.

1.1 History and evolution:

ONGC was established on 14th August 1956 by government of India for development and exploration of domestic oil and natural gas to reduce dependence on imported energy. Initially it was formed as a commission and later converted to a statutory body in 1959 and subsequently became oil and natural gas corporation limited in 1994.

The discovery of Mumbai high oil field in 1997 was a major milestone in ONGC's growth as it increased India's domestic oil production and transformed the country's energy sector.

1.2 Areas of operation:

ONGC covers different stages of oil and natural gas value chain:

1.2.1) Exploration- It is the initial stage of ONGC's operations. It involved identifying potential crude oil and natural gas reserves by geological and geophysical studies, seismic surveys and drilling activities to determine the presence of hydrocarbon. It is done in both offshore and onshore regions to identify commercially viable reserves for future extraction and production activities.

1.2.2) Production- Production involves the extraction of crude oil and natural gas from identified hydrocarbon reserves using drilling equipment and production facilities on both offshore and onshore fields in India. The extracted resources are processed and transported for further refining and distribution in different plants located all over the country.

1.2.3 Offshore operations- ONGC manages several major offshore oil and gas fields which contribute significantly to India's domestic hydrocarbon production. These operations involve exploration, drilling, extraction and maintenance activities using specialized equipment in marine environments. Offshore operations require advanced technologies, strict safety standards due to complex working conditions. Mumbai High is ONGC's most important offshore asset and has played a crucial role in increasing India's crude oil production.

1.2.4 Refining and petrochemicals- ONGC participates in refining and petrochemical activities through subsidiaries, partnerships and associated companies. Refining converts crude oil into usable products such as petrol, fuel, aviation fuel and lubricants, while petrochemical operations produce raw materials used in industries such as plastics, fertilizers and chemicals. These activities add value to the extracted hydrocarbons and support various industrial sectors.

1.2.5 International operations- ONGC has expanded its presence globally through **ONGC Videsh Limited(OVL)**. It undertakes exploration and production projects in several countries to strengthen India's energy security and diversify energy resources. International operations enable ONGC to access global hydrocarbon reserves and expand its oil and gas industry.

1.2.6 Sustainability initiatives- In response to changing energy trends and environmental concerns, ONGC is increasingly investing in renewable energy and cleaner technologies. It focuses on reducing environmental impact and support use of low carbon energy resources. This includes investments in renewable energy projects, energy efficiency improvements and environmentally sustainable operational practices.

2 Supply chain management:

Supply chain management refers to the planning coordination and management of activities involved in the flow of materials, information and services from sourcing raw materials to delivering final products to consumers. The primary objective of SCM is to ensure that products and services reach the right place at the right time and in the required quantity while minimizing costs and maximizing efficiency and enhancing overall performance of the company.

SCM comprises of several functions such as procurement, inventory management, production planning, logistics, warehousing, transportation and distribution.

2.1 Supply chain management at ONGC

In the oil and gas industry, supply chains are highly complicated due to large scale operations, specialized equipment requirement, strict safety regulations and geographically dispersed production sites.

At ONGC, the SCM plays an important role in supporting exploration, production, processing and distribution activities. The SCM ensures smooth movement of materials, equipment and information required for uninterrupted operations across the oil and gas value chain.

2.2 Supply chain flow at ONGC: From production to consumption :

The supply chain begins with exploration and extraction of crude oil and natural gas and continues until energy products reach industries and consumers. The process comprises of the following stages-

2.2.1) Exploration- The first stage involves finding potential crude oil and natural gas reserves through geographical surveys, seismic studies and drilling operations. Exploration activities are done both on offshore and onshore regions to locate hydrocarbon deposits.

2.2.2) Extraction and Production- Once the reserves are discovered, crude oil and natural gas are extracted using drilling equipment and production facilities. The extracted sources are collected from oil fields and offshore platforms for further processing.

2.2.3) Processing and refining -Extracted crude oil and natural gas undergo processing to remove impurities and convert them into usable products. Crude oil is refined into products

like petrol, diesel, aviation fuel and lubricants and natural gas is processed for commercial and industrial use.

2.2.4) Storage and transportation-Processed products are stored temporarily in warehouses and transported through pipelines, tankers or other logistic systems to distribution points.

2.2.5) Final consumption- Petroleum products and natural gas are utilized by industries, businesses, transportation sectors and households for energy generation and daily use.

3. Role of SAP at ONGC

3.1 Introduction to SAP ERP:

SAP(System, Applications and Products in Data Processing) is an Enterprise Resource Planning(ERP) software that integrates different business functions within an organization into a single unified system. It enables departments of an organization to share in real time, automate processes, improve coordination and support informed decision making.

3.2 Role of SAP at ONGC:

At ONGC, SAP serves as an integrated platform connecting key operational and administrative functions. ONGC operates over multiple locations and manages extensive procurement, production, finance and workforce activities. SAP plays a critical role in standardizing the process, ensuring efficient operations, maintain records and improve transparency across departments.

Major roles of SAP at ONGC are:

- Supply chain management(SCM)
- Material and inventory management
- Finance and accounting management
- Human Resource Management(HR)
- Plant maintenance and asset management
- Reporting and decision support
- Production and Billing Operations

4) Role of SAP in Procurement at ONGC:

Procurement is one of the important functions in SCM at ONGC as uninterrupted availability of materials, equipment and services is essential for exploration, production, maintenance and operational activities. SAP is used to integrate and automate procurement activities, ensuring transparency, traceability and efficient coordination among departments.

4.1) Procurement process in SAP at ONGC:

Requirement Generation/NTA → Purchase Requisition (PR) → Approval → Vendor Selection → Purchase Order (PO) → Goods Receipt (GR) → Invoice Verification → Payment Processing

4.1.1) Material Requirement Planning(MRP) and Requirement Identification-

The procurement process starts with identifying material requirements by analyzing

inventory levels, consumption patterns, maintenance schedules and operational demands to determine future material requirements.

MRP helps in:

- Identifying shortages of materials.
- Estimating procurement quantities.
- Supporting production and maintenance activities.
- Preventing stock-outs and operational interruptions.

The system generates procurement requirements based on existing stock and future needs.

4.1.2) Event generation and demand creation :

After identification of material shortages or operational requirements, SAP generates procurement related **events** or demand triggers. These events may arise due to:

- Inventory reaching reorder levels
- Maintenance requirements
- Production needs
- Project demands
- Operational emergencies

In ONGC, supporting documentation called NTA(Nodal Technical Agency) is generated for specialized procurement demands under which work centers put their demand collectively.

4.1.3 Purchase requisition (PR) Generation:

Following requirement identification and approvals, a **Purchase requisition(PR)** is created in SAP. PR is an internal request for procuring materials or services. A PR includes:

- Material description
- Quantity required
- Delivery location
- Required delivery date
- Estimated cost
- Department details

Sap records PRs and enables tracking throughout the procurement process.

4.1.4) Approval workflow :

PRs move through multiple approval levels depending on procurement value and hierarchy. SAP automates approvals and maintains records or authorizations.

4.1.5) Vendor selection and Purchase Order(PO) creation:

After approval, vendors are selected through tendering procedures, quotation evaluations or existing contracts. Based on approved procurement decisions, SAP generates a Purchase Order(PO) specifying:

- Vendor details
- Material specifications
- Quantity and price
- Delivery and price
- Delivery schedule
- Payment terms

The PO is the official procurement document issued to suppliers.

4.1.6) Good's Receipt (GR):

After delivery by suppliers, a **Good's receipt (GR)** is recorded in SAP to confirm delivery of materials according to PO. Inventory records are updated automatically following GR posting.

4.1.7) Invoice Verification:

The supplier submits an invoice after delivery and SAP verifies invoice details comparing :

Purchase Order (PO) → Goods Receipt (GR) → Invoice

This ensures consistency before payment approval.

4.1.8) Payment Processing:

After invoice verification, payment is processed through SAP's finance modules. The system tracks payment status and maintains financial records.

5) **Warehouse Management at ONGC:**

As ONGC handles variety of critical materials, spare parts, equipment and consumables, and effective warehouse management system is required for maintaining uninterrupted exploration and production operations. Warehouse management plays a crucial role in ONGC's supply chain management system as it ensures efficient receipt, storage, movement and issuance of materials required for production and operational activities. SAP and EWM(Extended warehouse management) help improve inventory visibility, storage efficiency and material traceability.

5.1) Warehouse Management process at ONGC:

5.1.1) Purchase Order and Supplier Dispatch:

The warehouse management process begins after a purchase order (PO) is issued to an approved supplier through SAP. Based on PO specifications, the supplier prepares materials for dispatch and arranges transport to designated ONGC warehouse or operational location.

5.1.2) Advanced Shipping Notification:

Before the physical shipment reaches the warehouse, the supplier sends an **Advanced Shipping Notification (ASN)** through warehouse management system. ASN contains shipment information such as material details, quantity, packaging information, transportation details and expected delivery date.

The ASN enables warehouse personnel to plan receiving activities in advance, allocate storage space, schedule inspections and prepare resources for unloading. This improves coordination between suppliers, procurement teams and warehouse staff, enhances inventory visibility and reduces receiving delays.

5.1.3) Inbound Delivery and Material Receipt:

Upon arrival of shipment, an **Inbound Delivery** is created and verified in SAP. Warehouse personnel compares received materials with the PO and ASN details to ensure accuracy before acceptance. This process ensures authorized and correctly delivery materials enter the warehouse system.

5.1.4) Quality Control(QC) and Inspection:

After the receipt, materials undergo **Quality Control(QC)** inspection by the inspection personnel to verify that they meet the required technical standards and examine the materials for defects, damage or deviations from procurement requirements.

Materials that pass QC are approved for warehouse storage while non-conforming materials are reported for corrective action or supplier follow-up.

5.1.5) Good's Receipt(GR):

Once inspection is completed and materials are accepted, a **Good's receipt(GR)** transaction is posted in SAP. GR updates inventory records and confirms that the order materials have been received.

5.1.6) Putaway and storage management:

Following GR, materials are moved to designated storage locations through a process called **Putaway**. Extended Warehouse Management(EWM) helps determines suitable storage locations.

Materials are stored in specific storage bins, each identified through unique bin location codes. Proper storage management enables easy material identification, retrieval and inventory control.

5.1.7) Inventory Monitoring and Warehouse Operations:

The warehouse needs continuous monitoring of inventory levels and material movements which is done through SAP. Storekeepers are responsible for maintaining stock accuracy, recording transactions and ensuring proper storage conditions.

ONGC utilize technologies such as **RFID tags, QR codes, bar-code system and automation tools** to improve accuracy, reduce manual efforts and enhance operational efficiency.

5.1.8) Material Issue and Consumption:

When operational departments require materials, an **Issue Requisition** is generated. Based on approved requirements, materials are issues from the warehouse and recorded in SAP. The issued materials are utilized and SAP automatically updates inventory records to reflect material consumption.

5.1.9) Stock Transfer Order(STO):

In situations where materials need to moved between warehouses or operational locations, a Stock Transfer Order(STO) is created in SAP. STO facilitates the controlled movement of materials while maintaining traceability and stock visibility across locations.

5.1.10) Outbound Delivery:

The final stage of warehouse management operations consists of **Outbound Delivery**, where materials are picked, packed and dispatched from warehouse to user departments, project sites, drilling locations or other ONGC facilities. Necessary documentation is generated in SAP and material movement can be tracked until delivery date.

Purchase Order → ASN → Inbound Delivery → Quality Inspection → Goods Receipt (GR) → Goods Receipt Voucher (GRV) → Putaway → Storage Bin → Inventory Management → Issue Requisition / STO → Outbound Delivery → Consumption

6) Logistics Management at ONGC

Logistics management at ONGC's supply chain operations, ensures efficient movement of materials, equipment, spare parts and consumables between warehouses, operational bases, ports, offshore installations and remote operational sites.

ONGC utilizes SAP and the Offshore Logistics Management System(OLMS) to manage transportation, delivery planning, inventory movement, vessel scheduling, shipment tracking and offshore logistics. It provides visibility, traceability and coordination throughout the logistics network.

6.1) Logistics Management Process at ONGC

6.1.1) Material Requirement and Logistics Planning :

The logistics process begins when operational departments raise requirements for materials, equipment or consumables. The logistics team verifies material availability in warehouse and site stores through SAP and plans the movement of materials based on operational priorities and delivery requirements.

The objective is to ensure the required materials are available at the right location and at the right time to support various activities carried out at ONGC.

6.1.2) Stock Transfer Order(STO):

When materials need to be transferred between warehouses, operational locations, offshore bases or site stores, a **Stock Transfer Order(STO)** is created in SAP.

The STO authorizes the movement of materials between storage locations while maintaining inventory visibility and traceability. It allows ONGC to monitor stock movement across different facilities and ensures proper inventory control throughout the transportation process.

6.1.3) Shipping Point Determination:

After STO creation, the appropriate shipping point is determined based on the destination and transportation requirement. The shipping point is responsible for coordinating dispatch activities, shipment preparation and transportation planning.

6.1.4) Shipping Conditions and Delivery Planning:

The shipping conditions are assigned on the basis of the final destination of the materials.

Materials are classified for:

➤ Onshore Delivery

➤ Offshore Delivery

Based on shipping conditions, the logistics team plans transportation mode, packaging requirements, loading schedules and delivery priorities. Proper planning helps ensure safe and timely delivery of materials to operational locations.

6.1.5) Outbound Delivery:

Once materials are ready for dispatch, an **Outbound Delivery Document** is created in SAP. Warehouse personnel pick, pack and prepare materials according to delivery requirements. Necessary shipping documents are generated and transportation arrangements are finalized before dispatch.

Outbound Delivery serves as the formal authorization for material movement from the warehouse to the destination location.

6.1.6) Goods Issue(GI) and Post Goods Issue(PGI)

After the shipment is dispatched, a **Post Good's Issue(PGI)** transaction is executed which confirms materials have officially left the warehouse and are in transit.

The GI and PGI updates inventory records, shipment status and logistics documentation while providing real time visibility of materials movement.

6.1.7) Offshore Logistics Management System(OLMS):

Following PGI, delivery information is processed through Offshore Logistics Management System(OLMS) which is a specialized logistics platform used by ONGC for planning, coordinating and monitoring offshore logistics activities. This system is integrated with SAP and provides visibility into shipment status, cargo allocation, vessel scheduling, voyage planning and delivery priorities.

OLMS enables efficient utilization of logistics resources and supports timely delivery of materials to offshore installations.

6.1.8) Consignment and Transportation Management:

Materials are grouped into consignments and assigned to suitable transportation modes based on delivery requirements

Transportation may be carried through:

- Road transport
- Supply vessels
- Specialized logistics carrier
- Helicopter services for urgent requirements

Managing consignments properly helps improve shipment tracking, transportation efficiency and delivery coordination.

6.1.10) Vessel Operations and Voyage Planning:

For offshore operations, materials are transported through supply vessels operating between logistics bases and offshore installations.

Vessel Voyages are planned by considering factors such as material priority, vessel capacity, weather conditions, operational schedules and offshore demand.

Proper voyage planning ensures optimum utilization of vessels and uninterrupted support to offshore production and drilling activities.

6.1.11) GPS Tracking and Remote Logistics Monitoring:

GPS based tracking systems are used in ONGC to enhance visibility and control over logistics operations.

It enables logistics personnel to monitor shipment locations, track vessel movements, verify delivery progress and coordinate transportation activities in real time.

This is particularly important for remote offshore locations where continuous monitoring is required to ensure operational reliability.

6.1.12) Goods Receipt (GR) and Delivery Confirmation

Upon successful delivery at the destination, a **Goods Receipt (GR)** transaction is posted in SAP.

GR confirms the receipt of materials and updates inventory records at the receiving location.

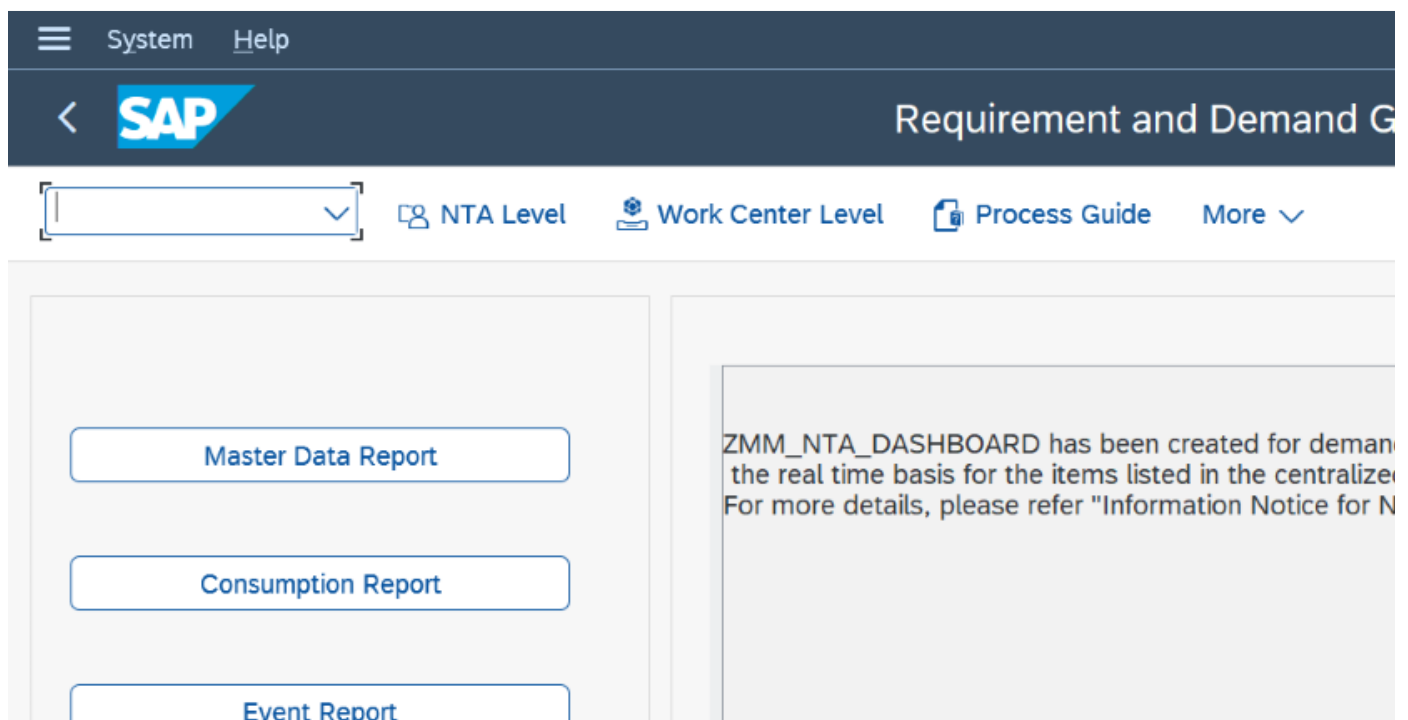
Material Requirement → STO Creation → Shipping Point Determination → Shipping Condition Assignment → Customs Clearance (if applicable) → Outbound Delivery → Goods Issue (GI) → Post Goods Issue (PGI) → OLMS → Consignment Creation → Vessel / Road → Voyage Planning → GPS Tracking → Goods Receipt (GR) → Material Consumption

Case Study

Problem Statement- Till 2023, procurement activities were decentralized, with individual work centers working independently procuring materials and maintaining records through physical files. This led to fragmented information, duplication of efforts, difficulty in tracking procuring activities and challenges in consolidating data across departments. The lack of an integrated system affected coordination among centers and limited visibility of material and procurement information. There, ONGC required a centralized digital platform to streamline procurement processes, improve data management, automate processes and enhance organizational coordination.

Solution Implemented:

To overcome the limitation of the decentralized procurement and manual record keeping, ONGC adopted SAP as an integrated enterprise resource planning system along with the Need for Tender Approval(NTA) for tender management. The implementation of these systems transformed the procurement process by centralizing procurement activities, standardizing procedures, improved transparency and enables seamless coordination among various departments. Through SAP and NTA, procurement activities are executed through a structured workflow beginning with identification of material requirements and ending with receipt of materials and payment to vendors.



Procurement at ONGC using SAP and NTA

1. Identification of material requirement: The procurement process begins when the concerned department reviews the requirement and determines whether procurement is necessary based on available inventory, consumption patterns and operational priorities. Requirement arises for a material, spare part, equipment or consumable item is identified.

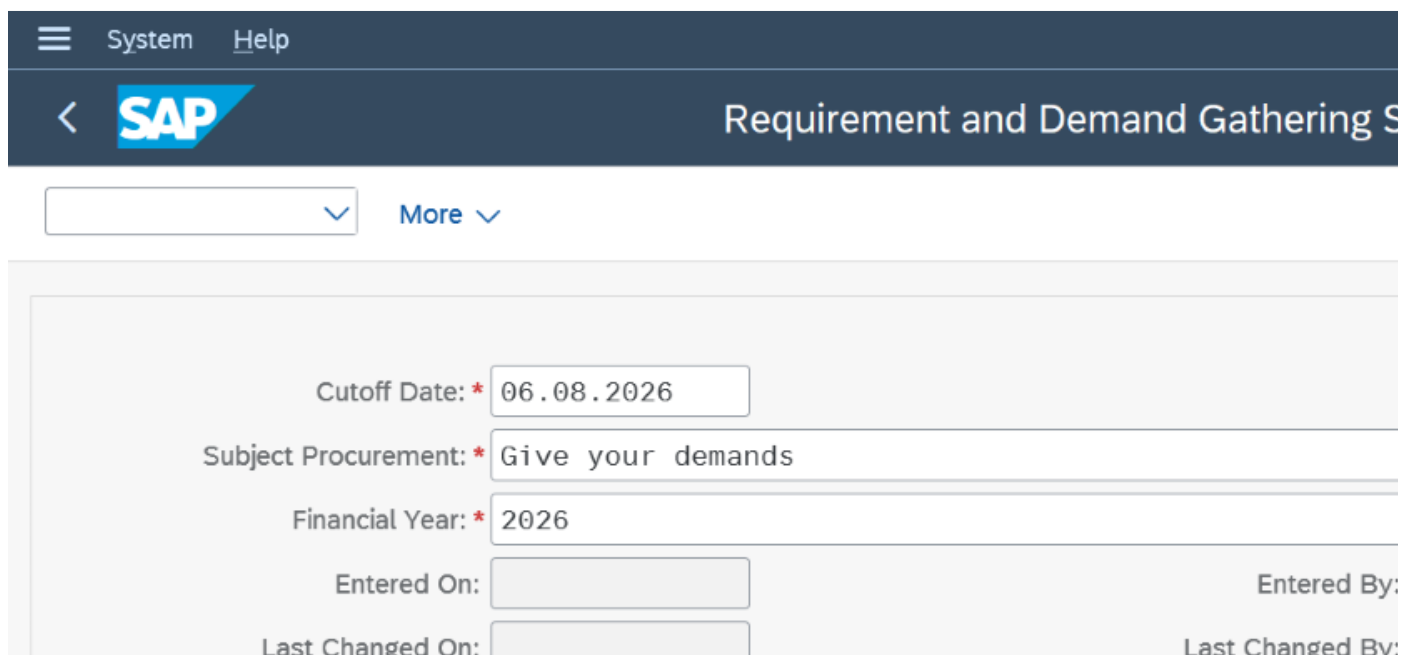
The requirement may arise due to operation needs, maintenance activities, project execution, stock replenishment or anticipated future consumption.

2. Creation of Procurement Event by NTA: After the requirement is established , the NTA creates a procurement event in SAP which serves as a common procurement platform through which multiples work centers can place their demands.

During event creation the following details are entered into the system:

- Material description and specifications
- Material group and category
- Estimated quantity
- Procurement Type
- Delivery requirements
- Event validity period
- Purchasing organization details
- Other tender related information

Upon creation, the system generated a unique Event Number, which becomes the reference point for all subsequent procurement activities.



The screenshot shows the SAP 'Requirement and Demand Gathering S' form. The header includes a menu icon, 'System', and 'Help'. Below the header is a navigation bar with a back arrow, the SAP logo, and the title 'Requirement and Demand Gathering S'. A search bar with a dropdown arrow and a 'More' link with a dropdown arrow are located below the navigation bar. The main form area contains several input fields with red asterisks indicating required fields:

- Cutoff Date: * 06.08.2026
- Subject Procurement: * Give your demands
- Financial Year: * 2026
- Entered On: [empty field]
- Entered By: [empty field]
- Last Changed On: [empty field]
- Last Changed By: [empty field]

Icon	Plant	Plant Description	Demand Doc	Status	Cl
	10E2				

Information

Event Document E000000276 created.

3. Publication and Availability of the event:

Once approved, the procurement event becomes available in SAP for relevant work centers. The event acts as a centralized procurement mechanism where work centers require the same material can participate. Work centers utilize the common event to submit their requirements. This ensures standardization of procurement activities and enables integration of organizational demand.


Requirement and Demand Gathering System

Add Attachments
List Files
More

Cutoff Date: 06.08.2026

Event Document Number: E000000276

Event Doc Release Status

Subject Procurement: GIVE YOUR DEMANDS

Financial Year: 2026

Entered On: 04.06.2026

Entered By:

Last Changed On:

Last Changed By:

Released On:

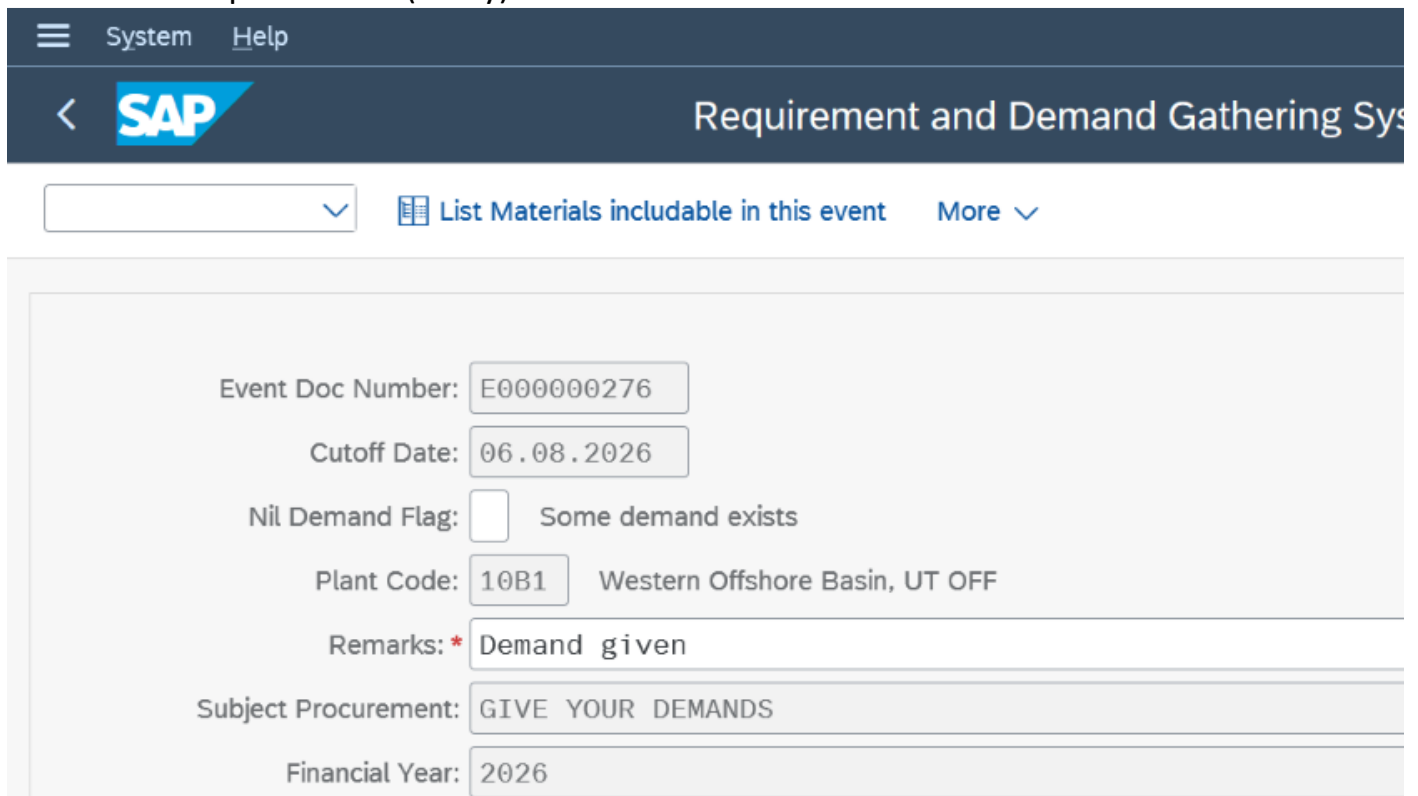
Released By:

4. Purchase Requisition(PR) creation by work centers:

After procurement event is released by NTA, individual work centers assess their requirements and create PRs in SAP against the event. The PR serves as a formal request for procurement containing details as:

- Event number
- Material description
- Required quantity
- Delivery location
- Required delivery date
- Account assignment details

➤ Technical specification(if any)



The screenshot shows the SAP Requirement and Demand Gathering System interface. At the top, there is a navigation bar with 'System' and 'Help' links. Below this, the SAP logo is visible on the left, and the title 'Requirement and Demand Gathering Sys' is on the right. A search bar is present, followed by a button labeled 'List Materials includable in this event' and a 'More' dropdown menu. The main content area displays a form with the following fields:

Event Doc Number:	E000000276
Cutoff Date:	06 . 08 . 2026
Nil Demand Flag:	☐ Some demand exists
Plant Code:	10B1 Western Offshore Basin, UT OFF
Remarks: *	Demand given
Subject Procurement:	GIVE YOUR DEMANDS
Financial Year:	2026

5. Consolidation of PRs:

SAP consolidated the PRs raised by various work centers against the same event. Procurement authorities can view the total requirement generated across all participating work centers and proceed with procurement based on the aggregate quantity.

6. Procurement Processing and Tendering

Following consolidation, the procurement authority proceeds with the procurement process which includes:

- Preparation of tender documents
- Vendor identification
- Tender publication
- Receipt of bids
- Technical evaluation
- Commercial evaluation
- Recommendation of a successful bidder
- Approval from competent authority.

All stages are recorded and monitored through SAP, ensuring transparency and compliance with procurement guidelines.

7. Purchase Order Creation:

Upon completion of the evaluation process and finalization of the supplier, a **Purchase Order(PO)** is created in SAP.

The PO contains:

- Vendor details
- Material specifications

- Ordered quantities
- Delivery schedules
- Pricing information
- Payment terms
- Delivery locations

The PO serves as a formal document between ONGC and the selected supplier.

8. Supply and Delivery of Materials:

The supplier dispatched the materials according to terms specified in PO.

Materials are transported to the designated receiving locations or warehouses ensuring a timely delivery

Upon arrival, materials are received by the concerned receiving authority.

9. Goods receipt in SAP:

After physical verification of the delivered materials, a **Good's Receipt (GR)** is recorded in SAP.

During this stage:

- Quantity received is verified
- Material condition is inspected
- Delivery documents are checked
- Receipt details are entered into the system

The GR updates inventory records and confirms that the ordered materials have been received.

10. Distribution to concerned work centers:

Once goods are received and inspected, materials are issues or allocated to the respective work centers based on the demands raised against the event.

Procurement Event → Demand Creation → Purchase Order → Goods Receipt → Material Issue

Lack of End-to-end Procurement Visibility:

Although the NTA dashboard provides information related to procurement activities, it does not offer complete end-to-end visibility of the entire procurement lifecycle on a single screen. Often difficulties are faced in tracking progress of a material from demand generation to final receipt.

To cater this end-to-end visibility requirement, a report is generated in SAP with the help of standard SAP query program by linking various tables through their key fields. Thus consolidating data and providing a comprehensive view of the procurement process.

Query Development and Analysis:

1) Creation of User Group: In the standard query area, a user group was created in SQ03 named "zmm_nta_ad" to manage authorizations, organize reporting activities and link infosets with queries for controlled access.

Menu: User group Edit Goto Settings Environment System Help

< **SAP** User Groups: Initial S

Search: [] Additional functions More ▾

Query area: Standard Area (Client-specific)

User group: [zmm_nta_ad] []

[Change] [Create]

[Display] [Description]

[Assign users and InfoSets]

2) Development of infoSet: An infoSet was created in SQ02 named “zmm_nta_es” to establish relationship between the required database tables.

Menu: InfoSet Edit Goto Extras Settings Environment System Help

< **SAP** InfoSet: Initial Screen

Search: [] Trash Additional functions More ▾

Query area: Standard Area (Client-specific)

InfoSet: [ZMM_NTA_ES] []

[Change] []

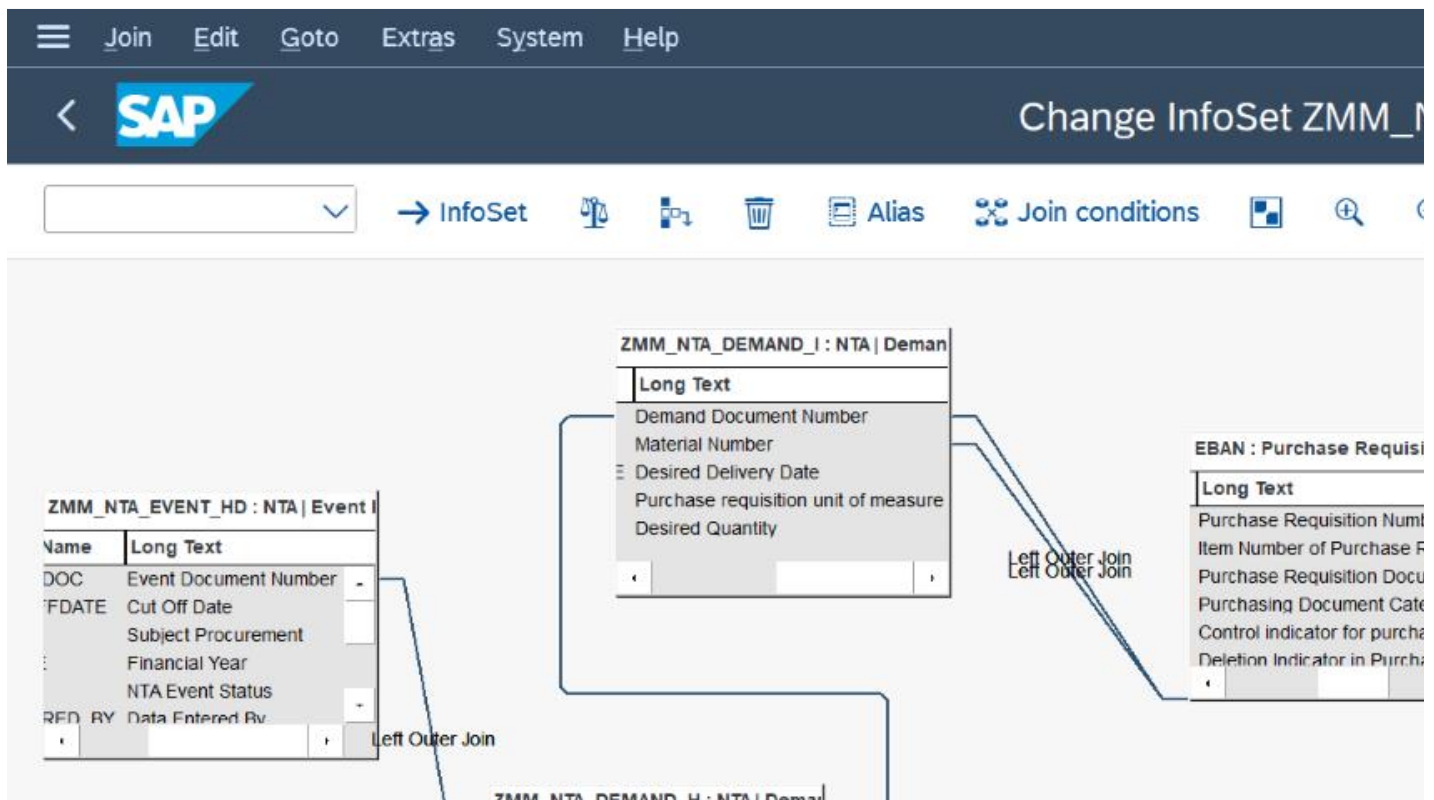
[Display] []

[Role/User Group Assign]

The tables joined are:

- Event header
- Demand header
- Demand item
- EBAN(Purchase Requisition)

➤ EKBE(History of Purchasing Document)



- I. Event Header to Demand Header table is linked using field of “Event Document Number” as it uniquely identifies an event. Every demand is created under a particular event, this field establishes the relationship between an event and its associated demands. These two tables are linked using **left outer joint** so as to display all the events even if no demand has yet been created for them. The event record will still appear in the query output.
- II. The demand header table is linked to demand item table using field of “Demand Document Number” as the demand document number uniquely identifies a demand. It connects the overall demand information stored in the header table with the individual material/item details stored in the item table. These two links are linked using **Inner joint** to ensure that only demands containing valid item details were included in report and preventing display of incomplete demand records in the final query.
- III. The demand item table is linked to the EBAN table by BANFN and BNFPO fields uniquely identifying the PR and its item enabling tracking from the demand item to the procurement document. They are connected through a left outer joint to display all demand items even if PR has not yet been created for them.
- IV. The EBAN and EBAN_00 alias has been linked as consecutive inner left joints cant be linked together. Also to ensure PR records remain available in the result set while extending the data model to subsequent tables.
- V. EBAN_00 and EKBE are linked through PR Number and Item Number using left outer joint, enables tracking the procurement status of each requisition item.

They are linked through left outer joints to display all the PR items even if no purchase order, Good's Receipt has yet been recorded in EKBE.

SAP

Query from User Group ZMM_NTA

Query area: Standard Area (Client-specific)

Query: Change

QuickViewer InfoSet Query Display De

Queries of user group ZMM_NTA_AD : query report

Name	Title	InfoSet	Logical Database	Table/View/Join	InfoSet Title

In SQ01 a query is created “zmm_nta_mp” to retrieve, analyze and present data from linked SAP tables in a structured report format.

SAP

Query ZMM_NTA_MP layout

Data fields	List fields	Selection F...	Technical Name	Field group
Table join	35	12		
> NTA Event Header Details	6	1	ZMM_NTA_EVENT_HD	
> NTA Demand Header Details	13	3	ZMM_NTA_DEMAND_H	
> NTA Demand Item Details	2	0	ZMM_NTA_DEMAND_I	
> Purchase Requisition	6	0	EBAN	
> Purchase Requisition	0	0	EBAN_00	
> History per Purchasing Document	0	0	EKBE	
> Additional Fields	5	5	---	
> Local additional fields	3	3		

Query Painter Edit Goto Tools Extras System Help				
Query ZMM_NTA_MP layo				
▼ Test Q Q+ Ruler All tools on/off Header Foo				
Data fields	List fields	Selection F...	Technical Name	Field gro
▼ NTA Event Header Details	6	1	ZMM_NTA_EVENT_HD	
Event Document Number	☐	☐	ZMM_NTA_EVENT_HD-ZMM_E...	NTA Ev
Cut Off Date	☒	☐	ZMM_NTA_EVENT_HD-ZMMC...	NTA Ev
Subject Procurement	☒	☐	ZMM_NTA_EVENT_HD-ZMMS...	NTA Ev
Financial Year	☒	☒	ZMM_NTA_EVENT_HD-ZMMS...	NTA Ev
NTA Event Status	☒	☐	ZMM_NTA_EVENT_HD-STATUS	NTA Ev
Data Entered By	☒	☐	ZMM_NTA_EVENT_HD-ZMM_E...	NTA Ev
Data Entered On	☒	☐	ZMM_NTA_EVENT_HD-ZMM_E...	NTA Ev
Data Last Changed By	☐	☐	ZMM_NTA_EVENT_HD-ZMM_CHANGED_B	
Data Last Changed On	☐	☐	ZMM NTA EVENT HD-ZMM ...	NTA Ev

From the basic list list fields and selection fields have been selected which will be displayed in the query.

Query Painter Edit Goto Tools Extras System Help				
Query ZMM_NTA_MP layo				
▼ Test Q Q+ Ruler All tools on/off Header Foo				
Data fields	List fields	Selection F...	Technical Name	Field gro
> NTA Demand Item Details	2	0	ZMM_NTA_DEMAND_I	
> Purchase Requisition	6	0	EBAN	
> Purchase Requisition	0	0	EBAN_00	
> History per Purchasing Document	0	0	EKBE	
> Additional Fields	5	5	---	
▼ Local additional fields	3	3		
number of days for event release	☒	☒	%Z_0005	Purchas
E2D	☒	☒	%Z_0006	NTA E
E2DR	☒	☒	%Z_0007	NTA E
< >				

Some local additional fields have also been selected which were created through calculations and are displayed in the query. They are done to enhance the analytical capability of the report, local calculation fields were created in SAP query.

▼

🏠

📄

🔍

⚙️

Basic List

Statistics

Ranked List

More ▼

Fields

	Short name	Local
NTA Event Header Details:		
☐	Event Document Number:	
☒	Cut Off Date:	
☒	Subject Procurement:	
☒	Financial Year:	
☒	NTA Event Status:	
☐		

▼

🏠

📄

🔍

⚙️

Basic List

Statistics

Ranked List

More ▼

Fields

	Short name	Local
☐	Data Released On:	
☒	E2D:	☒
☒	E2DR:	☒
NTA Demand Header Details:		
☒	Demand Document Number:	
☒	Event Document Number:	

Menu: Query Edit Goto Extras Settings Environment System Help

SAP Change Query ZMM_NTA_M

Fields

	Short name	Local
☒	Text: Status:	
☒	Created By:	
☒	Created On:	C
☐	Data Last Changed By:	
☐	Data Last Changed On:	
☐	Flag:	
☒	MRP Released By:	

Menu: Query Edit Goto Extras Settings Environment System Help

SAP Change Query ZMM_NTA_M

Fields

	Short name	Local
☒	Data Concurred On:	
☒	Data Approved By:	
☐	Data Approved On:	D
NTA Demand Item Details:		
☐	Demand Document Number:	
☒	Material Number:	
☐	Text: Material Number:	

Four data fields were first selected from the Event and demand tables and assigned short names for use in calculations.

- **A-** Data Entered On(Event Entry Date)
- **B-** Data Released on(Event Release Date)
- **C-** Created on(Demand Creation Date)

- **D-Data Approved On(Demand Approval date)**

Using the data fields, the following calculated fields were created:

1. Number of Days for Event release(E2RELEASE):

The screenshot shows the 'Field definition' form for a new field named 'E2RELEASE'. The form includes the following fields:

- Short Name:** * E2RELEASE
- Field Description:** number of days for event release
- Heading:** number of days for event release
- Field group:** Purchase Requisition

Below the field definition, there is a 'Properties' section with the following options:

- ☐ Same attributes as field
- ☐ Text field
- ☒ Calculation field
- ☐ Date Field

Additional properties include:

- No. of Characters:** (empty field)
- Number of Digits:** 10
- Decimal Plac** (empty field)

The screenshot shows the 'Define Field: Complex Calculation' form. It contains the following fields:

- Condition:** A<>0
- Formula:** B-A
- Condition:** (empty field)
- Formula:** (empty field)
- Condition:** (empty field)
- Formula:** (empty field)

Formula:

E2RELEASE=B-A

This calculation determines the number of days taken between the date and event was entered into the system and the date it was released.

2. Event to Demand Creation Time(E2D):

Field definition

Short Name: * E2D

Field Description: E2D

Heading: E2D

Field group: NTA | Event Header Details

Properties

☐ Same attributes as field

☐ Text field

☒ Calculation field

☐ Date Field

☐ Time Field

☐ Symbol

No. of Characters:

Number of Digits: 10

Decimal Place

Define Field: Complex Calculation

Condition A>0

Formula C-A

Condition

Formula

Condition

Formula

Formula:

E2D=C-A

This calculation measures the number of days between event creation and demand creation.

3. Event to Demand Approval:

Field definition

Short Name: *

E2DR

Field Description:

E2DR

Heading:

E2DR

Field group:

NTA | Event Header Details

Properties

☐ Same attributes as field

☐ Text field

No. of Characters:

☒ Calculation field

Number of Digits:

10

Decimal Places:

☐ Date Field

☐ Time Field

Define Field: Complex Calculation

Condition

A>0

Formula

D-A

Condition

Formula

Condition

Formula

Formula:

E2DR=D-A

This measures the number of days between event creation and demand approval.

For all calculations, condition **A>0** was applied to ensure calculations are performed only when a valid Event Entry Date is available. This prevents incorrect calculations due to missing data values.

These calculated fields allow user to monitor procurement timelines, identify delay and assess the efficiency of different stages of the procurement process.

SAP QUERY:

#	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Demand Document Number	Event Document Number	Plant	Nil Demand Flag	Created By	Created On	MRP Released By	MRP Released On	MIC Released By	MIC Released On	Concurred By	Concurred On	Approved By
2	D000000926	E000000103	21A1		132059	19-06-2024	132059	19-06-2024			68047	21-06-2024	71388
3	D000000926	E000000103	21A1		132059	19-06-2024	132059	19-06-2024			68047	21-06-2024	71388
4	D000000046	E000000074	50D1		122617	14-05-2024	122617	14-05-2024			82294	07-06-2024	70221
5	D000000046	E000000074	50D1		122617	14-05-2024	122617	14-05-2024			82294	07-06-2024	70221
6	D000000046	E000000074	50D1		122617	14-05-2024	122617	14-05-2024			82294	07-06-2024	70221
7	D000000046	E000000074	50D1		122617	14-05-2024	122617	14-05-2024			82294	07-06-2024	70221
8	D000001055	E000000096	25D1		134277	04-07-2024	134277	04-07-2024			92672	05-07-2024	71379
9	D000001055	E000000096	25D1		134277	04-07-2024	134277	04-07-2024			92672	05-07-2024	71379
10	D000001055	E000000096	25D1		134277	04-07-2024	134277	04-07-2024			92672	05-07-2024	71379
11	D000001055	E000000096	25D1		134277	04-07-2024	134277	04-07-2024			92672	05-07-2024	71379
12	D000001055	E000000096	25D1		134277	04-07-2024	134277	04-07-2024			92672	05-07-2024	71379
13	D000001055	E000000096	25D1		134277	04-07-2024	134277	04-07-2024			92672	05-07-2024	71379
14	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
15	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
16	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
17	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
18	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
19	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
20	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
21	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
22	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
23	D000000212	E000000073	50D1		106232	21-05-2024	106232	21-05-2024			82294	23-05-2024	83170
24	D000001024	E000000103	41A1		73569	27-06-2024	73569	08-07-2024			78411	08-07-2024	71545
25	D000001024	E000000103	41A1		73569	27-06-2024	73569	08-07-2024			78411	08-07-2024	71545
26	D000000014	E000000071	60D1		131416	11-05-2024	131416	11-05-2024			68779	15-05-2024	68636
27	D000000014	E000000071	60D1		131416	11-05-2024	131416	11-05-2024			68779	15-05-2024	68636

#	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC
1	Cut Off Date	Subject Procurement	Financial Year	Status	Entered By	Entered On	Material	Desired Quantity	Un Purch.Req.	Item	Document Type	Rel	Release	Req.date	NTA Event Statu	
2	08-07-2024	DMV, FCV & ETCC	2025-26	N	121820	12-06-2024	150013053		50 NO	1030114667	16	MPRJ	B	XXXXX	03-04-2025	New
3	08-07-2024	DMV, FCV & ETCC	2025-26	N	121820	12-06-2024	150013053		50 NO	1030114667	7	MPRJ	B	XXXXX	24-12-2024	New
4	29-07-2024	HP HOSES	2025-26	R	105118	08-05-2024	060033927		14 NO	1030114647	29	MPRJ	B	XXXXX	16-12-2024	Released
5	29-07-2024	HP HOSES	2025-26	R	105118	08-05-2024	060033881		12 NO	1030114647	23	MPRJ	B	XXXXX	16-12-2024	Released
6	29-07-2024	HP HOSES	2025-26	R	105118	08-05-2024	060000478		7 NO	1030114647	31	MPRJ	B	XXXXX	16-12-2024	Released
7	29-07-2024	HP HOSES	2025-26	R	105118	08-05-2024	060000002		13 NO	1030114647	26	MPRJ	B	XXXXX	16-12-2024	Released
8	29-07-2024	CEMENT ADDITIVES	2025-26	N	66484	30-05-2024	100000962	16,000.000	KG	1030113793	6	MPRJ			21-08-2024	New
9	29-07-2024	CEMENT ADDITIVES	2025-26	N	66484	30-05-2024	100000962	16,000.000	KG	1030113826	6	MPRJ	B	XXXXX	23-08-2024	New
10	29-07-2024	CEMENT ADDITIVES	2025-26	N	66484	30-05-2024	100111148	25.000	MT	1030113793	19	MPRJ			21-08-2024	New
11	29-07-2024	CEMENT ADDITIVES	2025-26	N	66484	30-05-2024	100111148	25.000	MT	1030113826	19	MPRJ	B	XXXXX	23-08-2024	New
12	29-07-2024	CEMENT ADDITIVES	2025-26	N	66484	30-05-2024	100101020	5.000	MT	1030113793	13	MPRJ			21-08-2024	New
13	29-07-2024	CEMENT ADDITIVES	2025-26	N	66484	30-05-2024	100101020	5.000	MT	1030113826	13	MPRJ	B	XXXXX	23-08-2024	New
14	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034012	400	NO	1030113562	37	MPRJ	B	XXXX	24-07-2024	Released
15	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034041	160	ST	1030113562	39	MPRJ	B	XXXX	24-07-2024	Released
16	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034044	341	NO	1030113562	40	MPRJ	B	XXXX	24-07-2024	Released
17	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034053	137	NO	1030113562	41	MPRJ	B	XXXX	24-07-2024	Released
18	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034018	1,040	NO	1030113562	38	MPRJ	B	XXXX	24-07-2024	Released
19	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	140008344	120	NO	1030113562	121	MPRJ	B	XXXX	24-07-2024	Released
20	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	061303612	260	NO	1030113562	81	MPRJ	B	XXXX	24-07-2024	Released
21	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034105	260	NO	1030113562	99	MPRJ	B	XXXX	24-07-2024	Released
22	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034080	120	NO	1030113562	80	MPRJ	B	XXXX	24-07-2024	Released
23	27-07-2024	CENTRALISER	2025-26	R	105118	08-05-2024	060034055	140	ST	1030113562	42	MPRJ	B	XXXX	24-07-2024	Released
24	08-07-2024	DMV, FCV & ETCC	2025-26	N	121820	12-06-2024	140006878		16 NO	1030114667	12	MPRJ	B	XXXXX	24-12-2024	New
25	08-07-2024	DMV, FCV & ETCC	2025-26	N	121820	12-06-2024	150013053		16 NO	1030114667	11	MPRJ	B	XXXXX	24-12-2024	New
26	30-06-2024	PROCUREMENT OF SHORT DRILL PIPES	2025-26	N	123735	08-05-2024	010001022		5 NO	1030115409	6	MPRJ	B	XXXXX	18-03-2025	New
27	30-06-2024	PROCUREMENT OF SHORT DRILL PIPES	2025-26	N	123735	08-05-2024	010103224		5 NO	1030115409	9	MPRJ	B	XXXXX	18-03-2025	New

1	NTA Event Status	Plant	Status	Release Indicator	Purchasing Group	number of days for event relea	E2D	E2DR		
2	New	Ankleshwar Asset, Ankleshwar	Approved	Fixed RFQ/purchase order			22	7	9	
3	New	Ankleshwar Asset, Ankleshwar	Approved	Fixed RFQ/purchase order			22	7	9	
4	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			82	6	30	
5	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			82	6	30	
6	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			82	6	30	
7	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			82	6	30	
8	New	Drilling Services, Cambay	Approved				49	35	36	
9	New	Drilling Services, Cambay	Approved	Fixed RFQ/purchase order			49	35	36	
10	New	Drilling Services, Cambay	Approved				49	35	36	
11	New	Drilling Services, Cambay	Approved	Fixed RFQ/purchase order			49	35	36	
12	New	Drilling Services, Cambay	Approved				49	35	36	
13	New	Drilling Services, Cambay	Approved	Fixed RFQ/purchase order			49	35	36	
14	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
15	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
16	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
17	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
18	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
19	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
20	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
21	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
22	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
23	Released	Drilling Services, Nazira	Approved	Fixed RFQ/purchase order			78	13	16	
24	New	Rajahmundry Asset, Rajahmundry	Approved	Fixed RFQ/purchase order			22	15	26	
25	New	Rajahmundry Asset, Rajahmundry	Approved	Fixed RFQ/purchase order			22	15	26	
26	New	Drilling Services,Agartala	Approved	Fixed RFQ/purchase order			28	3	8	
27	New	Drilling Services,Agartala	Approved	Fixed RFQ/purchase order			28	3	8	

The SAP Query was developed using an InfoSet that linked Event Header, Demand Header, Demand Item, Purchase Requisition and Purchase History Tables. Custom calculations were created to measure processing timelines between key procurement stages. The query generated a consolidated report, improving visibility and monitoring procurement activities.

Conclusion:

Supply Chain Management plays a vital role in ensuring the timely procurement and availability of materials required for organizational operations. At ONGC, the procurement process flows systematically from requirement identification and event creation to demand generation, approval and material procurement. The implementation of SAP has significantly improved process integration and coordination among different business operations. However, system demands continuous efforts in process simplifications, changes in validations as per changing business scenarios and adoption of new process for automation and digitization.

In line of above this case study has been taken up to examine a newly implemented business process “NTA DASHBOARD”. In this case study, it has been studied along with in requirement as per business process. A gap in reporting was identified and subsequently a query is developed to mitigate the non availability of end-to-end reporting.

Through this internship I gained practical exposure to SAP, procurement process, query development, data analysis and the application of Supply Chain concepts in a large public sector organization.

Acknowledgement:

I would like to express my sincere gratitude to ONGC for providing me with the opportunity to undertake this internship. I am deeply thankful to Mr. Ramandeep Singh Sir and Mr. Gaurav Nagpal Sir for their valuable guidance, support and encouragement throughout the internship period. I would also like to thank all the officials and staff members of ONGC who shared their knowledge and helped me gain practical exposure to Supply Chain Management and its practical operations through SAP. Their assistance contributed significantly to the successful completion of this report.